

(A) Training Architecture

Increasing spatial bandwidth

Visual sensor



Blind

non-visual baseline



Coarse

48^2 RGB
spatial collapse



Log-polar

256^2 + log-polar resampling
gaze-centred polar grid



Foveated

256^2 blur
peripheral blur



Uniform

256^2 RGB
full spatial layout

visual route

Encoder

LSTM

L0

L1

L2

Hidden State $\mathbf{H2}$

Freeze
&
Extract



Compass



Near-goal

non-visual sensors

π

Action

(B) Memory Analysis

$\mathbf{h}_2$ bank

500 ep $\times$ 5 cond.

Magnitude

what does $\mathbf{h}_2$ encode?

$\mathbf{h}_2 \xrightarrow{\text{linear / non-linear}} (x, y)$

Format (Spatial & Temporal)

how is it encoded?

$\angle(\mathbf{h}_2^x, \mathbf{h}_2^{x'}) \quad \mathbf{h}_2(t) \rightarrow \mathbf{h}_2(t')$

Consumption

what does the policy use?

$\tilde{\mathbf{h}}_2 = \text{ablate}(\mathbf{h}_2) \rightarrow \pi \rightarrow \Delta\text{SPL}$